If the switch in Fig. 7.10 opens at $t = 0$, find $v(t)$ for $t \ge 0$ and $w_C(0)$.

Practice Problem 7.2

Answer: $8e^{-2t}$ V, 5.333 J.

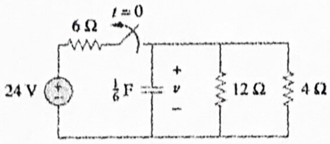


Figure 7.10
For Practice Prob. 7.2.

> **The Key to Working with a Source-Free *RC* Circuit Is Finding:**
> 1. The initial voltage $v(0) = V_0$ across the capacitor.
> 2. The time constant τ.

① $v(0^+) = v(0^-) = 8V$

② $R = 12 // 4 = 3 \Rightarrow \tau = RC = \frac{1}{2}$

$\therefore v(t) = 8e^{-2t}$ V

$w_C(0) = \frac{1}{2}Cv^2 = \frac{1}{2} \times \frac{1}{6} \times 8^2 = 5.333$ J

Find i and v_x in the circuit of Fig. 7.15. Let $i(0) = 7$ A.

Practice Problem 7.3

Answer: $7e^{-2t}$ A, $-7e^{-2t}$ V, $t > 0$.

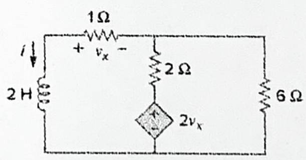
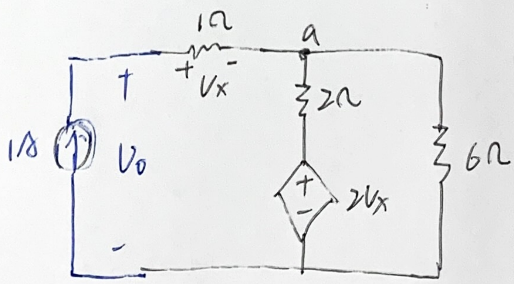


Figure 7.15
For Practice Prob. 7.3.

含受控源情况下求时间常数

① 找电感两端的等效电阻
有受控源，所以求等效电阻要施加激励求响应。



$v_x = 1$ V

对a点KCL: $1 = \frac{V_a - 2}{2} + \frac{V_a}{6}$

$\Rightarrow V_a = 3$

$\Rightarrow V_0 = 4 \Rightarrow R_{Th} = 4\Omega$

$\therefore \tau = \frac{L}{R} = \frac{1}{2}$

② 根据初始值和τ，写出表达式 $i = 7e^{-2t}$ A

$v_x = -i \cdot 1\Omega = -7e^{-2t}$ V

Practice Problem 7.10

Find $v(t)$ for $t > 0$ in the circuit of Fig. 7.44. Assume the switch has been open for a long time and is closed at $t = 0$. Calculate $v(t)$ at $t = 0.5$.

Answer: $(9.375 + 5.625e^{-2t})$ V for all $t > 0$, 11.444 V.

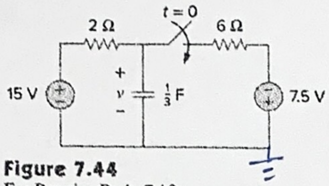


Figure 7.44
For Practice Prob. 7.10.

1. The initial capacitor voltage $v(0)$.
2. The final capacitor voltage $v(\infty)$.
3. The time constant τ.

① $v(0) = 15$ V

② $v(\infty) = -7.5 + \frac{15+7.5}{8} \times 6 = 9.375$ V

③ 电容两端电阻 $R = 2 \parallel 6 = \frac{3}{2}\ \Omega \Rightarrow \tau = RC = \frac{1}{2}$

$\therefore v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$

$= 9.375 + 5.625 \cdot e^{-2t}$ V

当$t=0.5$时 $v(0.5) = 11.442$ V

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Practice Problem 7.11

The switch in Fig. 7.47 is closed at $t = 0$. Find $i(t)$ and $v(t)$ for all time. Note that $u(-t) = 1$ for $t < 0$ and 0 for $t > 0$. Also, $u(-t) = 1 - u(t)$.

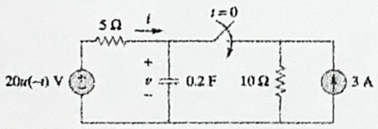

Figure 7.47
For Practice Prob. 7.11.

熟悉阶跃函数在电路描述中的应用

1. The initial capacitor voltage $v(0)$.
2. The final capacitor voltage $v(\infty)$.
3. The time constant τ.

Answer: $i(t) = \begin{cases} 0, & t < 0 \\ -2(1 + e^{-1.5t}) \text{ A}, & t > 0, \end{cases}$

$v = \begin{cases} 20 \text{ V}, & t < 0 \\ 10(1 + e^{-1.5t}) \text{ V}, & t > 0 \end{cases}$

① $v(0) = 20$ V ② $v(\infty) = 10$ V ③ $R = 5 \parallel 10 = \frac{10}{3} \Rightarrow RC = \frac{2}{3}$

$\therefore v = 10 + 10 \cdot e^{-1.5t}$ V

$\therefore v = \begin{cases} 10 + 10 \cdot e^{-1.5t} \text{ V} & t > 0 \\ 20 \text{ V} & t \le 0 \end{cases}$

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④ 根据v求i

$t<0$时, $i = 0$

$t>0$时, $i = -v/5 = -2 - 2e^{-1.5t}$ A

Practice Problem 7.13

Switch S_1 in Fig. 7.54 is closed at $t = 0$, and switch S_2 is closed at $t = 2$s. Calculate $i(t)$ for all t. Find $i(1)$ and $i(3)$.

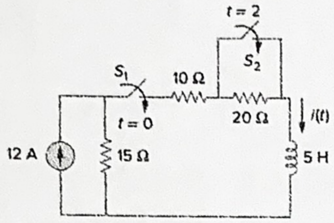

Figure 7.54
For Practice Prob. 7.13.

Answer:

$$i(t) = \begin{cases} 0, & t < 0 \\ 4(1 - e^{-9t}), & 0 < t < 2 \\ 7.2 - 3.2e^{-5(t-2)}, & t > 2 \end{cases}$$

$i(1) = 4$ A, $i(3) = 7.178$ A.

多开关多阶段分析

$i(0^-) = 0$

① 考虑 $0 < t < 2s$ 段.

$i(\infty) = 4$, $R = 45$ $\therefore L/R = \frac{1}{9}$

$\therefore i(t) = 4 + (0 - 4)e^{-9t}$

$= 4 - 4e^{-9t}$ A

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② 考虑 $t > 2s$ 段

$i(2^-) = 4 - 4e^{-18} = 4$

$i(\infty) = 12 \times \frac{3}{5} = 7.2$

$R = 25 \Rightarrow \tau = L/R = \frac{1}{5}$

$\therefore i(t) = 7.2 + (4 - 7.2)e^{-5(t-2)}$ A.

$= 7.2 - 3.2e^{-5(t-2)}$ A

③ $i(1) = 4 - 4e^{-9} = 4$ A

$i(3) = 7.2 - 3.2e^{-5} = 7.2$ A